

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 10%

Code of Conduct:

I RUTHIKA REDDY C 192421138 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ruthika Reddy

Rubrics for Evaluation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks(100)
<i>Understanding of Concepts</i>	25%	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>	
<i>Explanation</i>	25%	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>	
<i>Application</i>	20%	<i>Effectively applies concepts with strong</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>	

		<i>analytical ability</i>				
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>	
<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1. An image contains high-frequency noise. Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.

Answer

When an image contains **high-frequency noise**, the most appropriate frequency domain filter is the **Low-Pass Filter (LPF)**, particularly the **Gaussian Low-Pass Filter (GLPF)**.

A low-pass filter allows **low-frequency components** (which contain the main image information such as smooth regions and overall brightness) to pass through while attenuating or blocking **high-frequency components**, which mainly represent noise and fine details. The image is first transformed from the spatial domain into the frequency domain using the **Fourier Transform**. The filter is then multiplied with the frequency spectrum, and finally, the **Inverse Fourier Transform** is applied to reconstruct the filtered image.

The Gaussian Low-Pass Filter provides a smooth transition between passed and blocked frequencies, reducing ringing artifacts that occur with ideal filters.

Working

1. Convert the image to the frequency domain using FFT.
2. Apply the Gaussian Low-Pass Filter.

3. Suppress high-frequency noise.
4. Convert the filtered image back using Inverse FFT.

Impact on Image Quality

- Removes high-frequency noise effectively.
- Produces a smoother image.
- Slightly blurs edges and fine details.
- Preserves the overall structure of the image.

Justification

Since high-frequency noise exists in the higher frequency spectrum, suppressing these frequencies improves image quality while retaining the important low-frequency information.

2. You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect.

Answer

To preserve low-frequency components while removing noise, the **Butterworth Low-Pass Filter (BLPF)** is widely used. It provides a gradual transition between the passband and stopband, making it more effective than an Ideal Low-Pass Filter.

The Butterworth filter attenuates high-frequency components gradually instead of abruptly. This helps preserve important image information while reducing unwanted noise.

Working

1. Transform the image into the frequency domain.
2. Apply the Butterworth Low-Pass Filter.
3. Low frequencies pass without significant modification.
4. High-frequency noise is gradually reduced.
5. Perform the Inverse Fourier Transform.

Advantages

- Smooth transition reduces ringing artifacts.
- Better preservation of image details.
- Effective noise removal.
- Produces natural-looking images.

Application

Butterworth filtering is commonly used in medical imaging, satellite imaging, and digital photography where both noise reduction and image detail preservation are important.

Justification

Because the Butterworth filter gradually suppresses unwanted frequencies, it provides an excellent balance between smoothing and detail preservation.

3. An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.

Answer

For edge enhancement, a **High-Pass Filter (HPF)** is the most appropriate frequency domain filter. Edges correspond to rapid intensity changes, which are represented by **high-frequency components**.

A high-pass filter blocks low-frequency components while allowing high-frequency components to pass. As a result, edges become sharper and more visible.

Working

1. Convert the image into the frequency domain.
2. Apply the High-Pass Filter.
3. Remove low-frequency background information.
4. Retain high-frequency edge information.
5. Apply the Inverse Fourier Transform.

Effect on Frequency Components

- Removes low-frequency components.
- Enhances high-frequency components.
- Improves edge sharpness.
- May amplify high-frequency noise.

Applications

- Medical image enhancement.
- Fingerprint recognition.
- Satellite image analysis.
- Industrial inspection.

Justification

Since edges contain high-frequency information, enhancing these frequencies improves image sharpness and object visibility.

4. A smooth version of an image is required. Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.

Answer

Image smoothing is performed using a **Gaussian Low-Pass Filter** in the frequency domain. This filter reduces sudden intensity variations by removing high-frequency components while preserving low-frequency image information.

The Gaussian function provides a smooth frequency response, avoiding abrupt cutoffs that may introduce ringing effects.

Operation

1. Apply Fourier Transform.
2. Multiply the spectrum by the Gaussian Low-Pass Filter.
3. Remove high-frequency components.
4. Apply Inverse Fourier Transform.

Results

- Noise reduction.
- Smooth image appearance.
- Reduced sharp transitions.
- Slight edge blurring.
- Minimal ringing artifacts.

Applications

- Image preprocessing.
- Medical image enhancement.
- Face recognition.
- Noise removal before segmentation.

Justification

The Gaussian Low-Pass Filter provides effective smoothing while minimizing distortion, making it one of the most commonly used smoothing techniques.

5. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.

Answer

The most suitable segmentation technique is **Threshold Segmentation** because the object intensity is significantly higher than the background.

Thresholding converts a grayscale image into a binary image using a selected threshold value.

Working

- Select threshold value **T**.
- If pixel intensity $> T$, classify it as the object.
- If pixel intensity $\leq T$, classify it as the background.

Advantages

- Simple implementation.
- Fast computation.
- Effective for images with distinct brightness differences.
- Produces accurate segmentation.

Applications

- Medical image analysis.
- Industrial inspection.
- Document processing.
- Object detection.

Justification

Since the object is brighter than the background, threshold segmentation provides accurate and efficient separation.

6. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.

Answer

The appropriate method is **Global Thresholding**.

Global Thresholding uses one threshold value for the entire image to separate foreground from background.

Process

1. Read the grayscale image.
2. Select a threshold value.
3. Compare every pixel intensity with the threshold.
4. Assign pixels above the threshold to the foreground.
5. Assign pixels below the threshold to the background.
6. Generate a binary segmented image.

Advantages

- Easy implementation.
- Fast execution.
- Low computational complexity.
- Suitable for uniformly illuminated images.

Applications

- OCR systems.
- Medical image segmentation.
- Industrial quality inspection.
- Shape extraction.

Justification

Global Thresholding is effective when there is a clear intensity difference between the foreground and background.

7. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.

Answer

The **Sobel Edge Detection** method is widely used for detecting object edges.

The Sobel operator uses two convolution masks to calculate horizontal and vertical intensity gradients.

Working

1. Apply horizontal Sobel mask (G_x).
2. Apply vertical Sobel mask (G_y).
3. Compute gradient magnitude.
4. Threshold the gradient image.
5. Obtain the edge map.

Advantages

- Detects horizontal and vertical edges.
- Reduces sensitivity to noise.
- Easy implementation.
- Good edge localization.

Applications

- Face recognition.
- Medical imaging.
- Object detection.
- Machine vision.

Justification

The Sobel operator provides reliable edge detection while suppressing minor noise, making it suitable for most applications.

8. Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.

Answer

The most appropriate segmentation technique is **Region Growing**.

Region Growing starts with one or more seed pixels and expands the region by adding neighboring pixels having similar intensity values.

Working

1. Select seed pixel.
2. Compare neighboring pixels.
3. Add pixels with similar intensity.
4. Continue until no neighboring pixels satisfy the similarity condition.

Advantages

- Produces connected regions.
- Preserves object boundaries.
- Simple implementation.
- Effective for homogeneous regions.

Applications

- Medical image segmentation.
- Satellite image analysis.
- Brain tumor detection.
- Tissue classification.

Justification

Since neighboring pixels share similar intensity, Region Growing naturally groups them into meaningful regions.

9. Detected edges in an image appear broken. Explain and justify a method to improve edge continuity, including its role in post-processing.

Answer

The appropriate post-processing method is **Morphological Closing**, which consists of **Dilation followed by Erosion**.

Closing fills small gaps and connects broken edge segments without significantly changing object shapes.

Working

1. Apply Dilation to expand edge pixels.
2. Fill small gaps.
3. Apply Erosion to restore original edge thickness.
4. Obtain continuous edges.

Advantages

- Connects broken edges.
- Removes small holes.
- Improves edge continuity.
- Produces cleaner boundaries.

Applications

- Character recognition.
- Object tracking.
- Medical image processing.
- Shape analysis.

Justification

Morphological Closing improves edge connectivity, making object boundaries continuous and easier for further analysis.

10. You need to extract object outlines from an image. Describe and apply an appropriate technique, explaining how it identifies boundaries effectively.

Answer

The appropriate technique is **Boundary Extraction using Morphological Operations.**

Boundary extraction identifies only the outer pixels of an object by subtracting the eroded image from the original binary image.

Formula

$$\text{Boundary} = A - (A \ominus B)$$

Where:

- **A** = Original binary image
- **B** = Structuring element
- $\ominus$ = Erosion operation

Working

1. Convert the image into a binary image.
2. Apply erosion.
3. Subtract the eroded image from the original image.
4. The remaining pixels form the object's boundary.

Advantages

- Produces accurate object outlines.
- Removes interior pixels.
- Preserves object shape.
- Useful for shape analysis and recognition.

Applications

- Medical image analysis.
- Object recognition.
- Industrial inspection.
- Computer vision.

Justification

Boundary extraction effectively isolates the object's outline, making it suitable for feature extraction, shape analysis, and object recognition.